

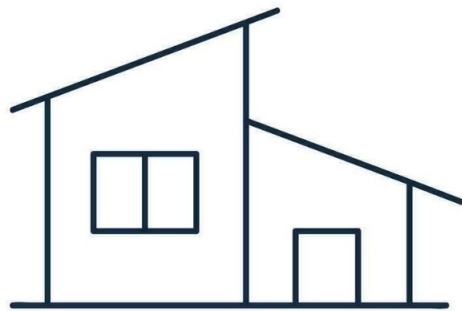
Forecasting Housing Outcomes with \$250k

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The \$250k Question: Buy or Rent?

\$250,000



- You have \$250,000 to invest
- Two choices: Buy a home or rent & invest
- Which of this path will be more profitable?
- That would vary by city, interest rate, mortgage rate, property tax and market in general

The Challenge: Making the Right Financial Decision



Interest Rates

Fluctuate, impacting affordability



Property Values

Vary by location, market trends



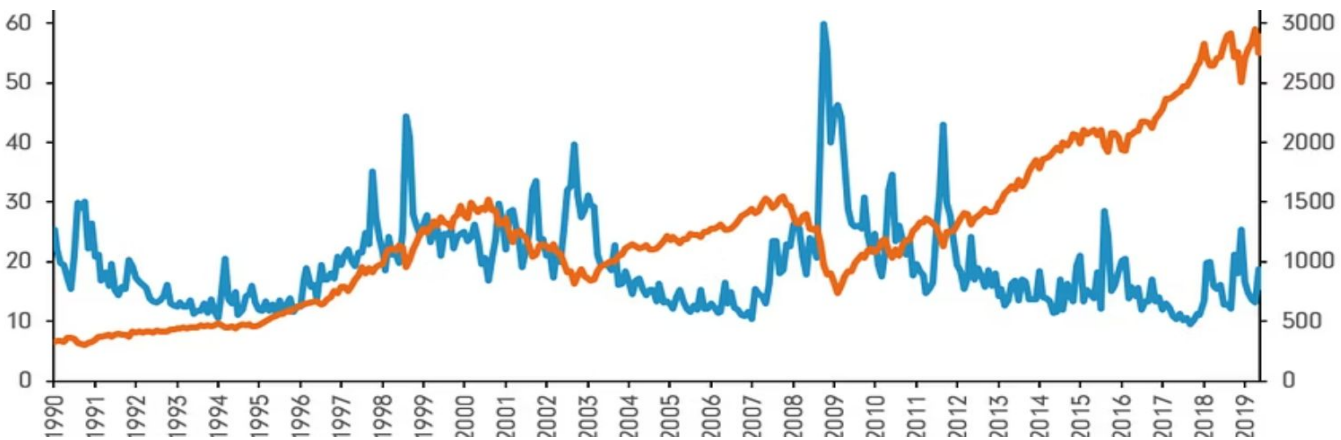
Rent Costs

Dynamic, influence budget

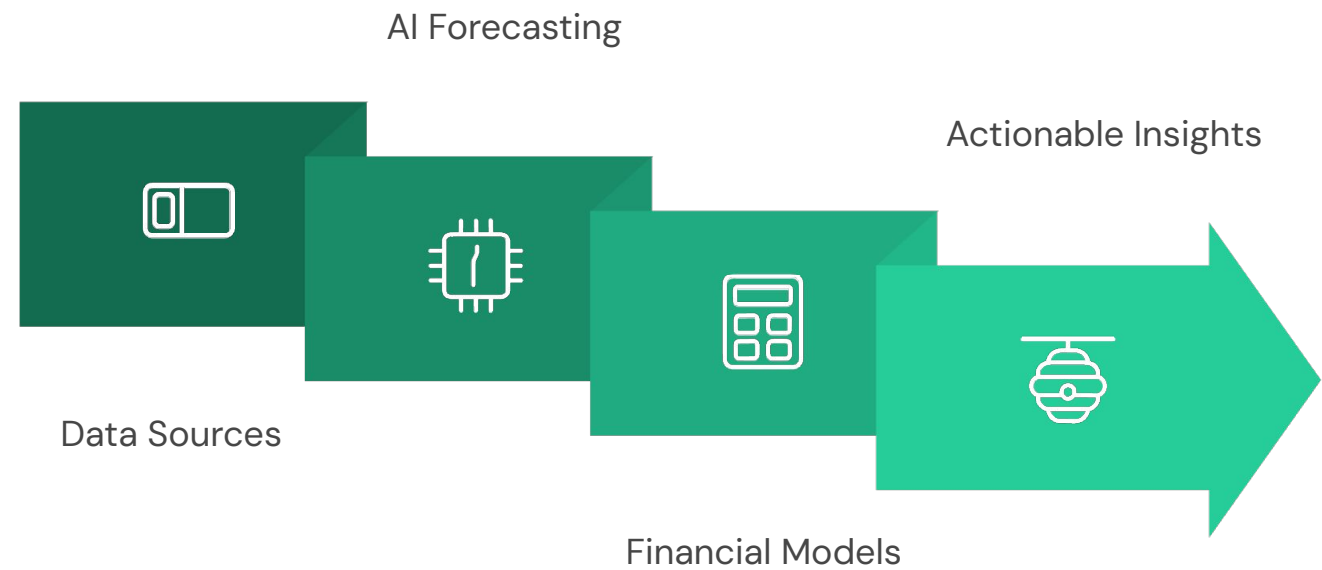


Investment Returns

Represents opportunity cost of capital



Our Data-Driven Approach



1

Data Sources

Zillow, FRED API for mortgage rates

3

Financial Modeling

NPV, IRR, break-even analysis

2

AI Forecasting

ARIMA models for prices & rates

4

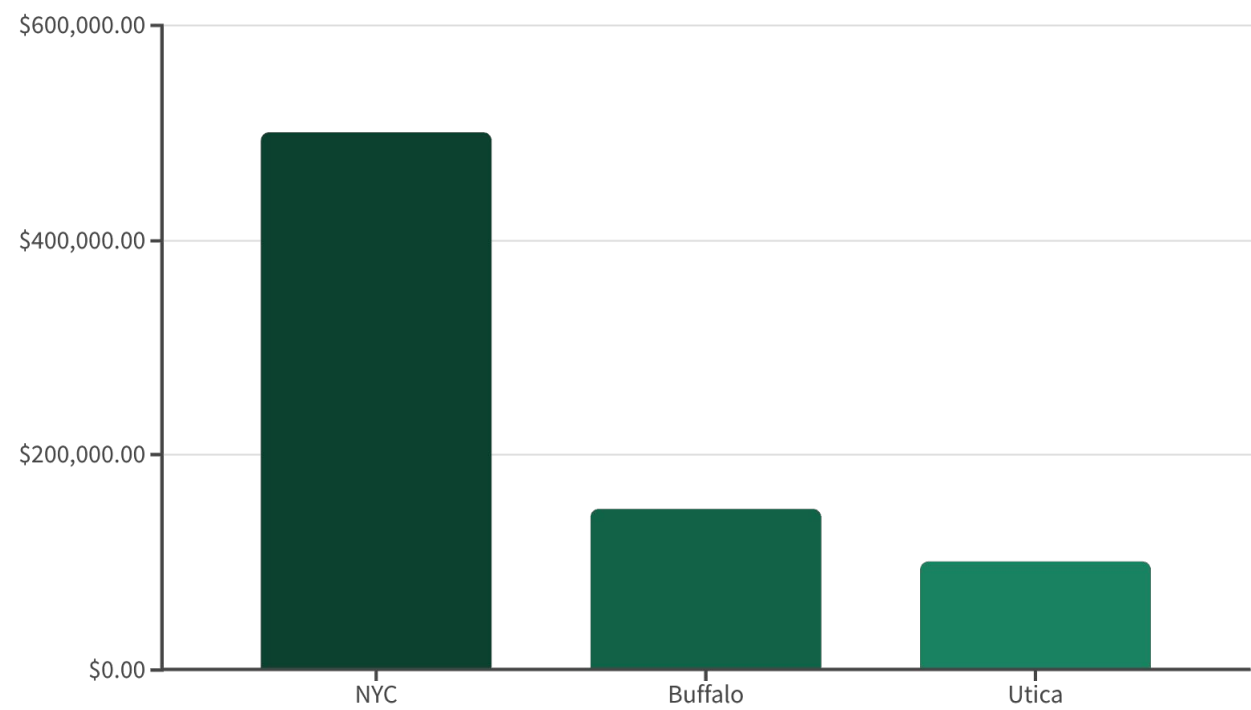
Multi-City Analysis

NYC, Buffalo, and Utica explored

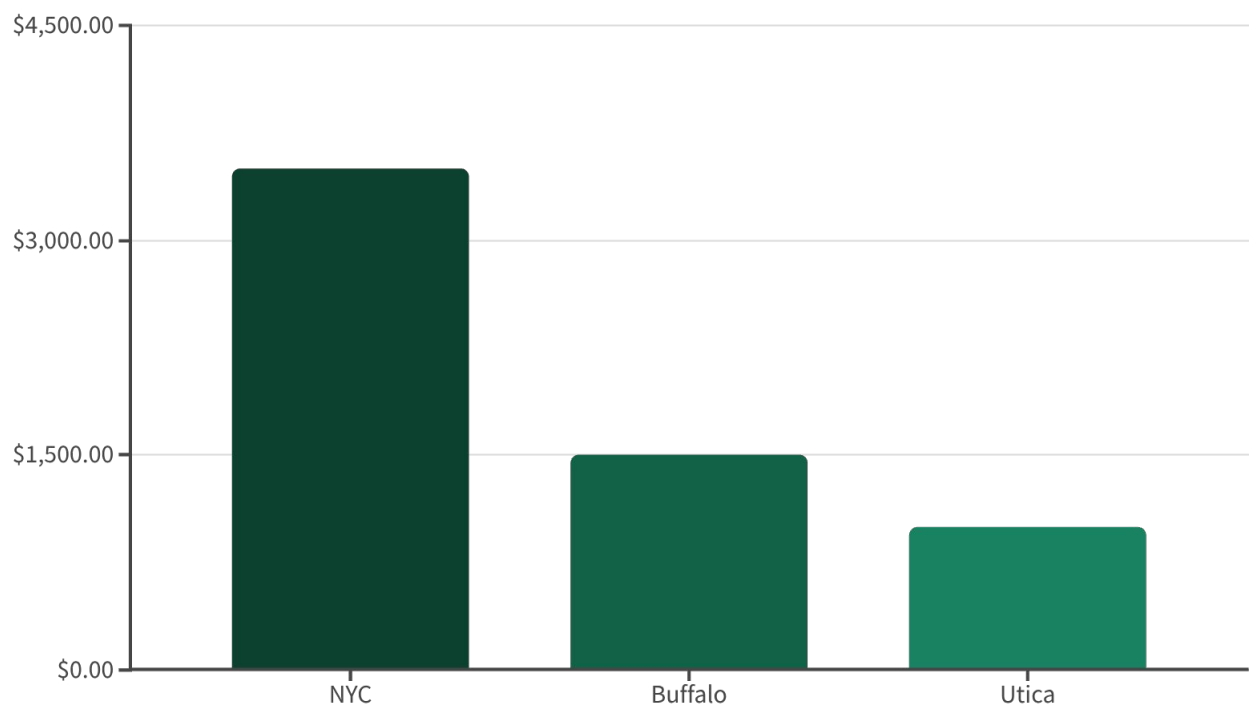
Current Market Conditions: What the Data Shows

Mortgage Rates: Around 6–7% (30-year fixed)

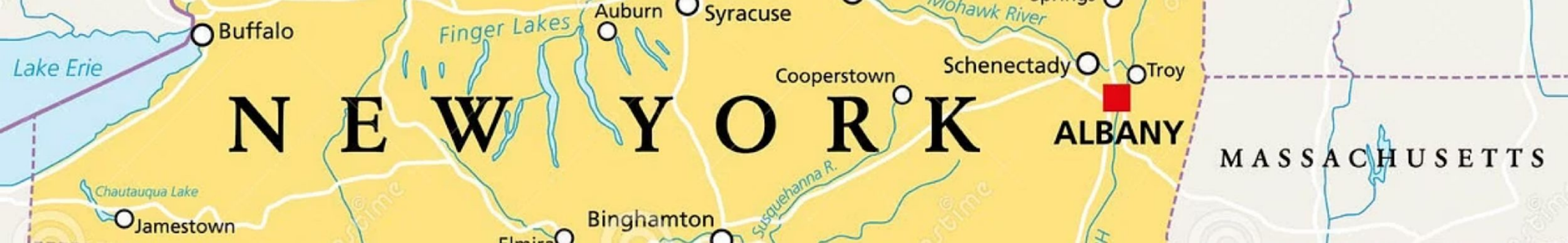
Property Values: Vary Dramatically



Rental Costs: Follow Similar Patterns



Market Volatility: Rates and prices change monthly, impacting buy vs. rent decisions.



The Answer Depends on Where You Live

1

NYC

Rent (\$69K savings, +\$180K wealth)

2

Utica

Rent (\$50K savings, +\$350K wealth)

3

Buffalo

Rent (\$56K savings, +\$364K wealth)

10-Year Horizon: Renting Wins in All Cities

→ 30-Year Horizon:

- NYC: Buying wins after 19.25 years
- Utica: Buying wins after 13.75 years
- Buffalo: Buying wins after 21.3 years

→ Key Takeaway:

Time horizon impacts decision more than city



When Does Buying Make Financial Sense?

Break-Even Point: Where cumulative buying costs = cumulative renting costs

1	NYC Month 231 (19.25 years) January 2045
2	Utica Month 165 (13.75 years) July 2039
3	Buffalo Month 256 (21.3 years) February 2047

Before Break-Even:

Renting is cheaper

After Break-Even:

Buying becomes cheaper & advantage grows

Wealth Projection: The Long-Term View

Why Buying Wins Long-Term

- Mortgage paid off
- House appreciation (243% forecast)
- Rent continues to rise (122% increase)

Why Renting Wins Short-Term

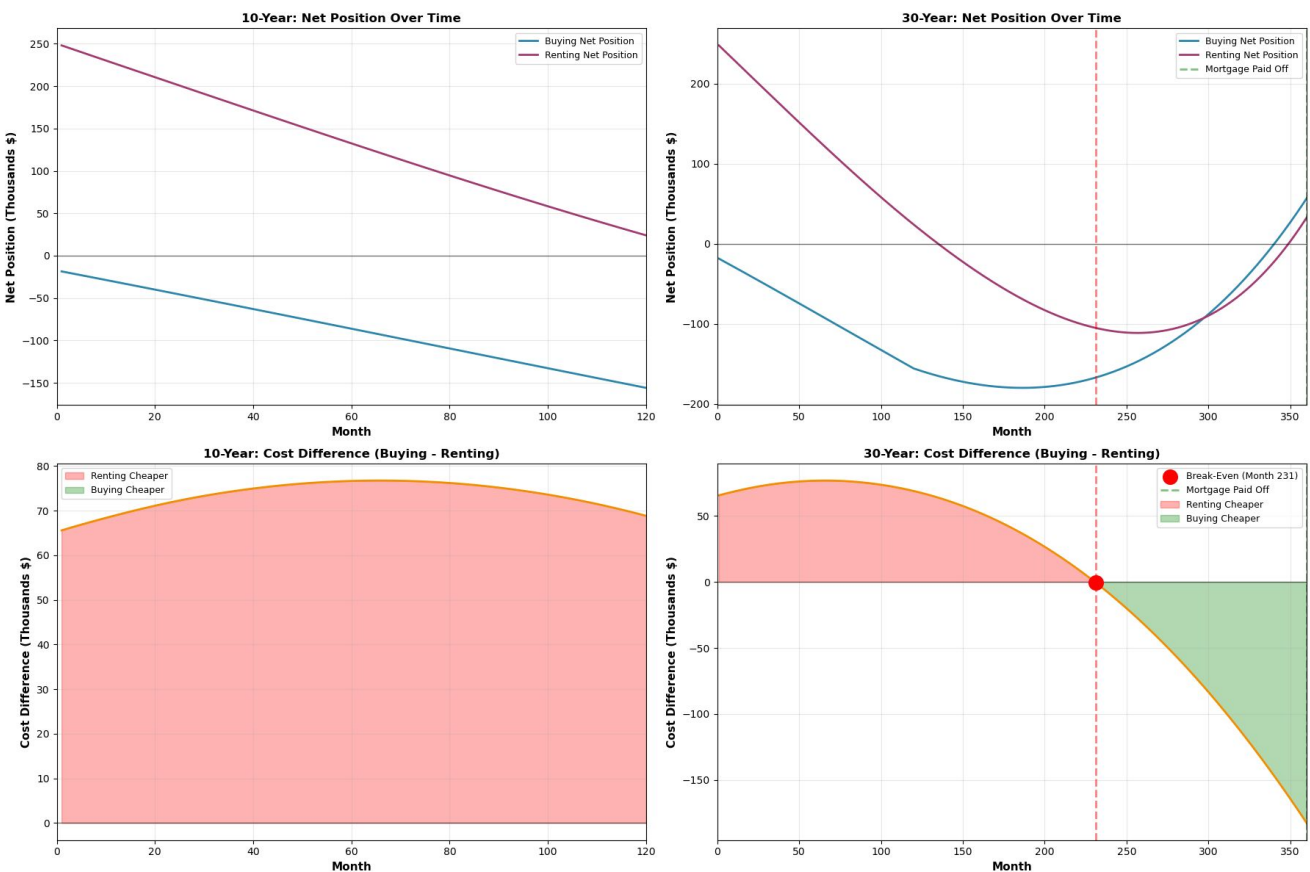
- Investment returns compound faster
- Lower upfront costs
- More flexibility

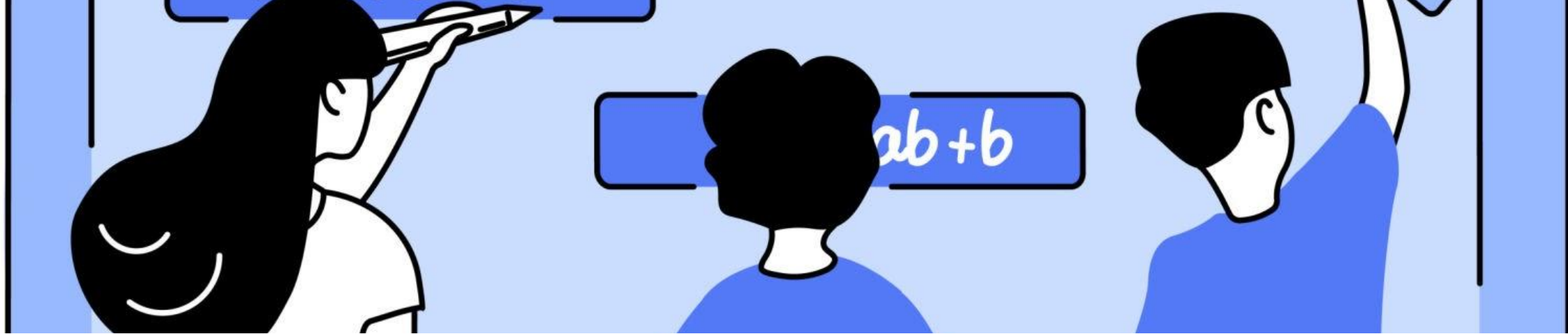
30-Year Final Wealth Comparison

NYC: Buying advantage of ~\$24K

10-Year Final Wealth Comparison

- NYC: Renting advantage of ~\$180K
- Utica: Renting advantage of ~\$350K
- Buffalo: Renting advantage of ~\$364K





Dashboard: Explore the Data Yourself



Real-Time Exploration



Visual Insights

Break-even, wealth, costs



Customizable Views

Compare cities side-by-side



Data-Driven Decisions

Insights backed by models

Accessible to non-technical users for informed decision-making.

Key Recommendations

Time Horizon Decision

Short-Term (0-13.8 years)

✓ RENT

Optimal for financial flexibility and avoiding large upfront costs associated with buying.

- Lower upfront investment
- Greater mobility and flexibility

Long-Term (13.8+ years)

✓ BUY

Ideal for building long-term equity and realizing significant financial benefits over time.

- Builds home equity
- Potential for long-term appreciation

The Bottom Line: Simple Decision Framework

 Moving within 10-15 years? → RENT

 Staying 15+ years? → BUY

Future Work

Multi-City Integration

- Expand to additional regions for broader market analysis.
- Incorporate break-even point comparisons across diverse locations.

Enhanced Features

- Integrate key economic indicators and tax implications.
- Develop robust neighborhood-level analysis capabilities.
- Interest rate impact modeling on affordability.
- Investment return projections and opportunity cost analysis.
- Rental income potential and passive income calculations.

User-Centric Tools

- Provide personalized buy/rent recommendations.
- Introduce interactive scenario planning and financial calculators.

